Toolkit: Chemistry of the Earth

Practical Six: Chemical Equilibrium: Thermodynamics and Phase Diagrams

Objectives:

- Familiarity with thermodynamic variables (enthalpy, entropy, Gibbs Free Energy) and the concept of equilibrium
- Construct a phase diagram from thermodynamic data
- Construct a phase diagram from experimental data
- Interpretation of basic phase diagrams and the Gibbs Phase Rule

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FEEDBACK

A. Thermodynamic variables (enthalpy, entropy, Gibbs Free Energy) and the concept of equilibrium

1. Two different minerals have the formula $CaCO_3$: calcite and aragonite. (These two minerals are *polymorphs* of $CaCO_3$.)

Thermodynamic data for calcite and aragonite are given in Table 1. We can use these data to determine which polymorph is stable at the Earth's surface (1 bar and 298.15 K).

Table 1: Thermodynamic data for minerals		
Mineral	ΔH^o_f kJ mol^{-1}	S^o J K^{-1} mol^{-1}
Calcite	-1207.3	92.68
Aragonite	-1207.21	90.21

ΔH^o_f is the enthalpy of formation of the mineral from the elements. The superscript "o" refers to the "standard state" which is 1 bar (0.1 MPa) and 298.15 K (25 °C).

S^o is the entropy of the mineral at the standard state reference conditions: 1 bar and 298.15 K.

We can write an equation for a reaction between calcite and aragonite:

$CaCO_3$ = $CaCO_3$
Calcite = *Aragonite*

The Gibbs free energy of the reaction (ΔG_r) is calculated from the equation:

$$\mathbf{\Delta G_r = \Delta H - T\Delta S}$$

Values for ΔH and ΔS are calculated as follows:

$\Delta H = \Delta H^o_f$(Aragonite) – ΔH^o_f(Calcite)

$\Delta S = S^o$(Aragonite) – S^o(Calcite)

(Note that these equations are right hand side minus left hand side for the mineral equation.)

a) Use the equations above to determine the Gibbs free energy of the reaction (ΔG_r):
Note that ΔH^o_f is given in units that include kJ, and S^o is given in units that include J, so you will need to take this into account. Also note that T must be in Kelvin. Include units for ΔG with your answer.

$\Delta H = -1207.21 - (-1207.3) = 0.09$ kJ mol^{-1}

$\Delta S = 90.21 - 92.68 = -2.47$ J K^{-1} mol^{-1} = -0.00247 kJ K^{-1} mol^{-1}

$\Delta G = \Delta H - T\Delta S = 0.09 - (298.15 * -0.00247) = 0.826$ kJ mol^{-1} or 826 J mol^{-1}

b) At equilibrium, the Gibbs free energy is equal on both sides of the equation ($\Delta G_r = 0$). A reaction proceeds in the direction that lowers the Gibbs free energy ($\Delta G_r < 0$).
Which of the two polymorphs of $CaCO_3$ is stable at the Earth's surface?

ΔG for this reaction is positive, so it will not proceed. Calcite is stable at the Earth's surface.

B. Construct a phase diagram from thermodynamic data

In question A above, we looked at the reaction between the two polymorphs of $CaCO_3$, calcite and aragonite. We will now investigate how the stability of these minerals changes with pressure and temperature.

Consider the same reaction as above:

$CaCO_3$ = $CaCO_3$
Calcite = *Aragonite*

Chemical equilibrium in this reaction is given by the equation:

$$P = 1 - (\Delta G_r/\Delta V_r)$$

where ΔV_r = V(aragonite) – V(calcite)

ΔG_r = ΔG(aragonite) – ΔG(calcite)

and values of ΔG for each mineral vary with temperature according to the Table below.

Table 3: Thermodynamic data for calcite and aragonite

	CALCITE	**ARAGONITE**	**Calculation of the equilibrium pressure**				
Temperature K	ΔG kJ mol^{-1}	ΔG kJ mol^{-1}	ΔG_r J mol^{-1}	ΔV J bar^{-1} mol^{-1}	P bar	P kbar	T °C
298.15	-1128.842	-1127.793	1049	-0.2784	3769	3.8	25
400	-1102.155	-1100.691	1464	-0.2784	5260	5.3	126.85
500	-1076.292	-1074.31	1982	-0.2784	7120	7.1	226.85
600	-1050.723	-1048.132	2591	-0.2784	9308	9.3	326.85
700	-1025.427	-1022.16	3267	-0.2784	11736	11.7	426.85
800	-1000.238	-996.251	3987	-0.2784	14322	14.3	526.85
900	-975.195	-970.468	4727	-0.2784	16980	17.0	626.85
1000	-950.299	-944.833	5466	-0.2784	19635	19.6	726.85

ΔV_r is the change in volume for the reaction. There is little change in the volume of calcite and aragonite with temperature or pressure, so we can assume that the (molar) volume of each mineral remains the same at each temperature:

V(calcite) = 3.6934 J bar^{-1} mol^{-1} V(aragonite) = 3.4150 J bar^{-1} mol^{-1}

(1 J = 10 cm^3 bar, so 1 J bar^{-1} mol^{-1} = 10 cm^3 mol^{-1}. Hence, V(calcite) = 36.934 cm^3 mol^{-1})

The value of ΔV_r (-0.2784 J bar^{-1} mol^{-1}) is given in the Table.

a) Complete the table above to calculate the pressure at which calcite and aragonite are in equilibrium, at temperatures of 298.15 K and 400 K. In the final column, convert the temperature from Kelvin to °C.

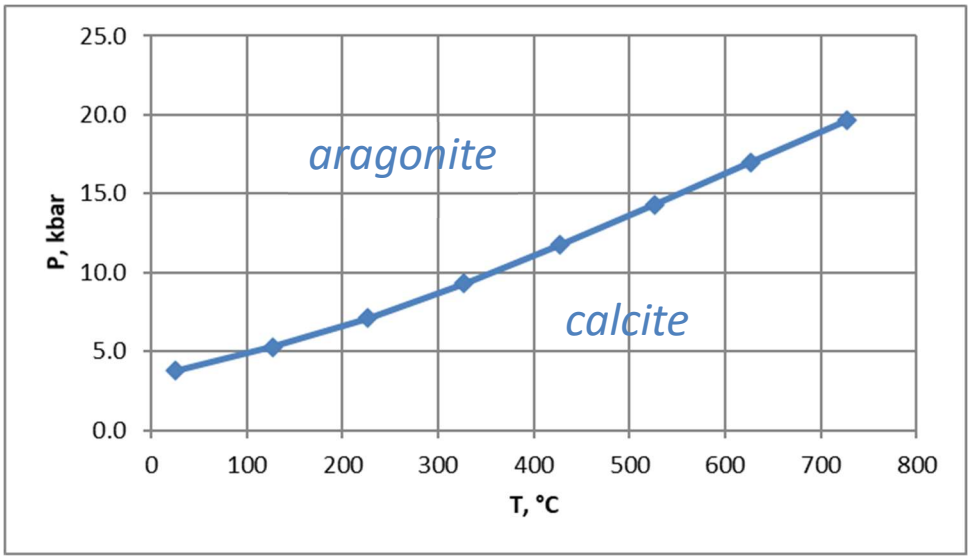
At 298.15 K:

ΔG_r = ΔG(aragonite) – ΔG(calcite) = -1127.793 – (-1128.842) = 1.049 kJ mol^{-1} = 1049 J mol^{-1}

$P = 1 - (\Delta G_r/\Delta V_r)$ = 1 – (1049/-0.2784) = 3769 bar = 3.769 kbar

T = 298.15 K = 298.15 – 273.15 = 25 °C

b) Plot the data from the table above in a graph of P (in kbar) on the y axis vs T (in °C) on the x axis. Draw a line that represents chemical equilibrium for the reaction calcite = aragonite. You have now drawn a phase diagram. The line is called a PHASE BOUNDARY. The regions each side of the line are STABILITY FIELDS for each mineral. Using the result of your calculation in question A, decide which mineral is stable in which field and label the diagram accordingly.



c) In general, which of the two minerals is more stable at higher pressure? How might this relate to the differences in the molar volumes of calcite and aragonite?

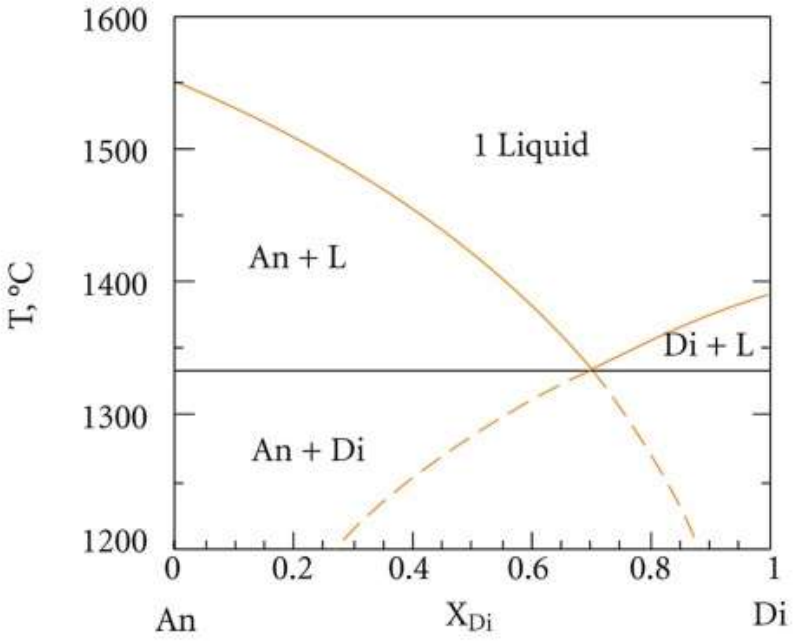
You have shown that calcite is the stable polymorph at the Earth's surface (25 °C and 1 bar pressure). Since this is below the line, calcite is the low-pressure polymorph and aragonite is the high-pressure polymorph. Calcite is stable under all conditions below the line, and aragonite is stable at all conditions above the line. If we are at conditions exactly on the line, both polymorphs are stable together.

The molar volume of aragonite is slightly smaller than the molar volume of calcite (see above). It is typical that a mineral phase with smaller volume will be more stable at higher pressure.

C. Construct a phase diagram from experimental data

Phase diagrams are also determined in experiments. The table below gives the results of experiments in which the two minerals, diopside and anorthite, were mixed together in different proportions and heated to different temperatures. The goal was to determine the melting relationships for the two minerals. The experiments were cooled quickly so that the phases present at the temperature of the experiment were preserved. Use the table of data to construct a phase diagram showing the phase boundaries and stability fields of anorthite, diopside and liquid.

In the lecture we looked at how this same phase diagram can be produced using thermodynamics. The calculated diagram is shown below. Your diagram should have the same general shape. However, the position of the eutectic, the minimum temperature of melting, will be different. Compare the two results for the eutectic and try to interpret why there is a difference. Which is more accurate, and why?



Phase diagram computed from thermodynamics

Eutectic:

X_{Di} = 0.7

T = 1334 °C

(Geochemistry: W. White)

Table 3: Diopside-Anorthite phase diagram		
Xdi	T	Phases present
0	1250	anorthite
0	1400	anorthite
0	1552	anorthite
0	1554	liquid
0.2	1270	anorthite + diopside
0.2	1275	anorthite + liquid
0.2	1450	anorthite + liquid
0.2	1500	liquid
0.5	1250	anorthite + diopside
0.5	1325	anorthite + liquid
0.5	1350	liquid
0.5	1500	liquid
0.8	1250	anorthite + diopside
0.8	1300	diopside + liquid
0.8	1350	diopside + liquid
0.8	1400	liquid
1	1270	diopside
1	1275	diopside
1	1391	diopside
1	1400	liquid

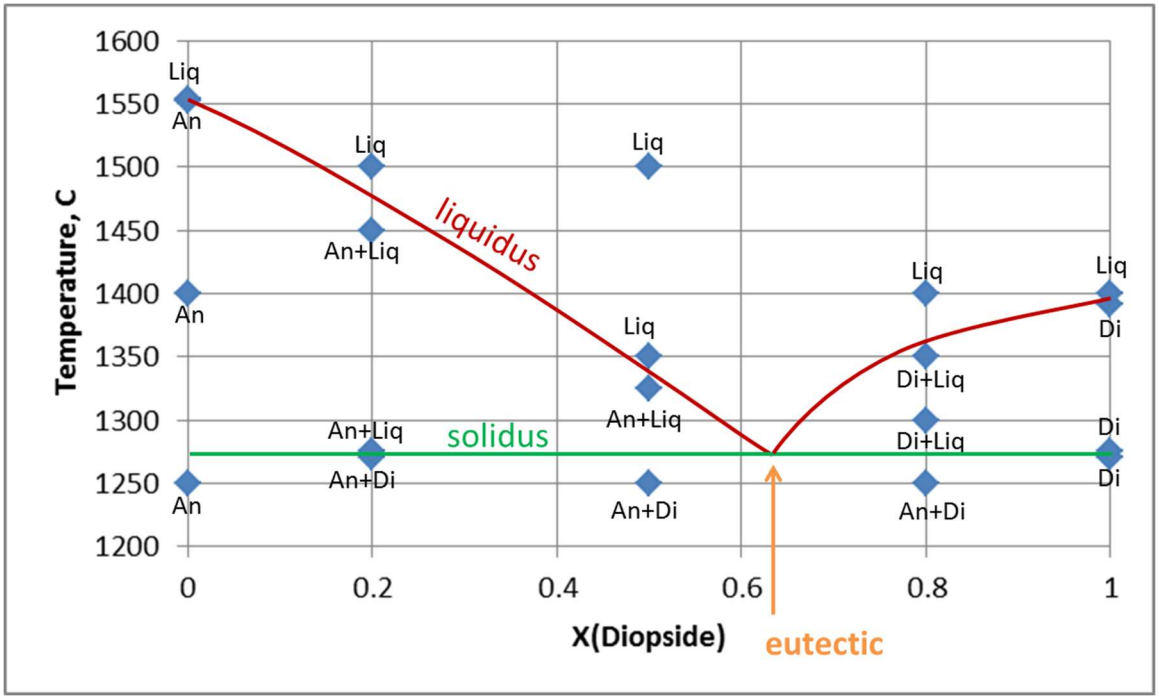
Phase diagram constructed from experiments

Eutectic:

X_{Di} = approximately 0.64

T = approximately 1272 °C (between 1270 and 1275)

Your diagram should look like the one below. Everything above the red line (the "liquidus") should be liquid, and everything below the green line (the "solidus") should be solid. The two regions between the red lines show stability fields for anorthite + liquid and diospide + liquid.



Hint: for each point, plot the point and label it with the phases present (An, Di, L)

Why is there a difference between the two diagrams? The thermodynamic calculation assumes that a mixture between anorthite and diopside is "ideal", in other words a liquid behaves just as if it contains separate An and Di components. In reality, bonding and interactions between atoms in the liquid mean that the liquid behaves differently. This is called "non-ideal" mixing and is the case for the real world. The experiments are a more accurate record of the real behaviour of the system.

D. Interpretation of basic phase diagrams and the Gibbs Phase Rule

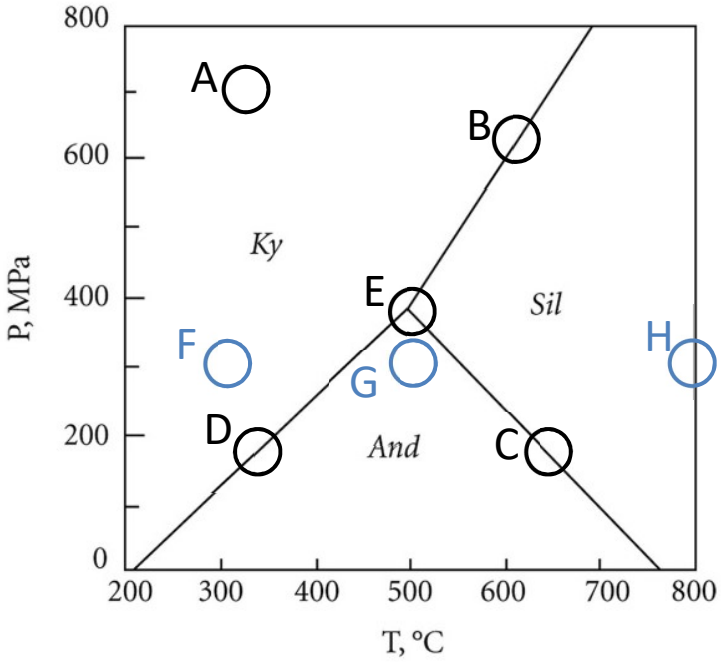
The Gibbs Phase Rule is: $f = c - \phi + 2$

c = components. The number of components in a system is the minimum number of chemical species required to describe the composition of all the phases in the system

ϕ = phases. A phase is a physically distinct body of matter that is (at least in principle) physically separable from the rest of the system.

f = degrees of freedom: the number of intensive variables that can be changed independently, without altering the state of the system, i.e. without changing the number of phases

1. The phase diagram below is for polymorphs of Al_2SiO_5, which are important minerals in metamorphic rocks. In this diagram, the composition is constant (Al_2SiO_5). There is only one component – Al_2SiO_5. The different polymorphs are kyanite, andalusite, and sillimanite.

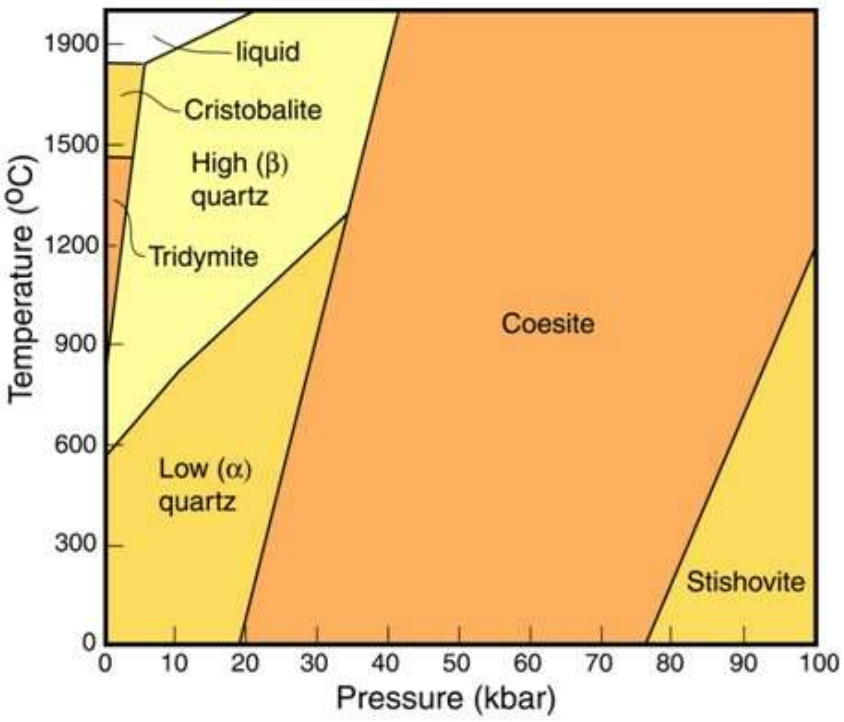


For each of the points in the phase diagram, determine the number of stable phases and calculate the degrees of freedom. You will need to plot points F, G, and H on the diagram.

Table 4: Gibbs Phase Rule

Point	T, P °C, Mpa	Stable Phases	Degrees of Freedom
A	xxxx	kyanite	2
B	xxxx	ky + sillim	1
C	xxxx	and + sillim	1
D	xxxx	ky + and	1
E	xxxx	ky + and + sillim	0
F	300, 300	kyanite	2
G	500, 300	andalusite	2
H	800, 300	sillimanite	2

2. The phase diagram below shows the stability of phases in the SiO_2 (silica) system.



(note that the number 1900 on the y axis should be 1800)

a) How many components are shown in this diagram? One – silica (SiO_2). Everywhere on the diagram the system has the same composition, SiO_2.

b) How many phases are shown in this diagram? Seven - each different coloured area is a different phase, including liquid.

c) List the polymorphs of SiO_2 shown in this diagram. Crsitobalite, tridymite, high quartz, low quartz, coesite, stishovite. (Liquid is a phase but not a mineral polymorph.)

d) Indicate on the diagram two points each where the degrees of freedom, f = 2, f = 1 and f = 0.

Any area on the diagram has f=2, any line has f = 1, and any junction between lines has f = 0 (e.g. the assemblage low-quartz – high quartz – coesite is stable at a single point, at which f = 0).

e) Indicate on the diagram where $\Delta G = 0$ for the reaction: high quartz = coesite

This means that both high quartz and coesite are stable, so it is the line between the two fields.

f) A silica mineral is found in a volcanic rock that erupts at 1200 °C. What polymorph would you expect for the silica mineral grains? (show this on the diagram)

Since the lava is erupting we can assume low pressure. At P = 0 and 1200 °C, the stable polymorph is tridymite.

g) A silica mineral is found in a sediment that formed at a pressure of 10 kbar and at 200 °C. What polymorph would you expect? (show this on the diagram)

This would be low quartz.

h) The same silica grain from the sediment in (g) is hit by a large meteorite impact and is converted into a different polymorph at a pressure of 70 kbar. What polymorph would you expect? (show this on the diagram)

This would be coesite. (The minerals coesite and stishovite were first identified in impact crater materials.)

i) At (approximately) what temperature does SiO_2 melt, at the surface of the Earth? (show this on the diagram)

At about 1700 °C (actually 1713 °C). This is indicated on the diagram on the y axis (P = 0), at the boundary between the fields of cristobalite and liquid.

j) If you find a grain of cristobalite, what kind of rock would it most likely have come from?

A volcanic rock – it needs very high temperatures.

k) How is it possible to find a grain of cristobalite at the Earth's surface (1 bar, 20 °C)?

The phase diagram shows the thermodynamically stable phases. However, many metastable or disequilibrium phases persist to surface conditions because reactions can only proceed with sufficient energy. *Kinetic effects* are very important in preserving disequilibrium minerals.